

A Unified Scalar Field for Cross-Sector Physical Anomalies: Forward–Backward Consistency Across Nuclear, Gravitational, Inertial, and Cosmological Domains

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Abstract

A wide range of physical anomalies—including nuclear decay-rate deviations, lunar gravitational MASCONE residuals, inertial navigation discrepancies, and cosmological scalar-field ratios—cluster in the same fractional deviation band of 10^{-4} to 10^{-3} . This paper presents a unified scalar field $G(t)$ that explains these anomalies through a single correction law $X_{\text{eff}} = X_0(1 + c_X G)$. We derive the scalar-field Lagrangian, field equation, stress–energy tensor, cosmological equation of state, collapse reversal condition, and relic formation mechanism. We demonstrate forward–backward consistency: given any $G(t)$, the theory predicts sector deviations; given sector deviations, the same $G(t)$ is reconstructed. This establishes the scalar field as a measurable physical quantity with cross-domain coherence.

1 Introduction

Multiple independent physical domains exhibit small but persistent anomalies in the fractional deviation band 10^{-4} – 10^{-3} . These include:

- nuclear decay-rate deviations (Mn-54, Si-32, Cl-36),
- lunar gravitational MASCONE anomalies,
- Apollo inertial navigation residuals,
- cosmological scalar-field ratios from DESI ($K/V = 0.0811$).

This paper proposes that these anomalies arise from a single scalar field $G(t)$ with amplitude $G \sim 10^{-3}$, coupled to physical sectors through a unified correction law. We show that the theory is internally consistent, empirically anchored, and invertible across all sectors.

2 Unified Correction Law

We postulate that all physical quantities X subject to small deviations follow:

$$X_{\text{eff}}(t) = X_0(1 + c_X G(t)), \quad (1)$$

where c_X is a sector-specific coupling constant.

The fractional deviation is:

$$\frac{\Delta X(t)}{X_0} = c_X G(t). \quad (2)$$

2.1 Sector Instantiations

Nuclear decay:

$$\frac{\Delta \lambda(t)}{\lambda_0} = \alpha G(t), \quad G(t) = \frac{\Delta \lambda(t)}{\alpha \lambda_0}. \quad (3)$$

Gravity:

$$\frac{\Delta g(t)}{g_0} = \gamma G(t), \quad G(t) = \frac{\Delta g(t)}{\gamma g_0}. \quad (4)$$

Inertia:

$$\frac{\Delta a(t)}{a_0} = \gamma G(t), \quad G(t) = \frac{\Delta a(t)}{\gamma a_0}. \quad (5)$$

These forward and inverse relations allow reconstruction of $G(t)$ from data.

3 Scalar-Field Lagrangian and Dynamics

We adopt the canonical scalar-field Lagrangian:

$$\mathcal{L}_G = \frac{1}{2} g^{\mu\nu} \nabla_\mu G \nabla_\nu G - \frac{1}{2} m_G^2 G^2. \quad (6)$$

The Euler–Lagrange equation yields:

$$\nabla_\mu \nabla^\mu G + m_G^2 G = 0. \quad (7)$$

In an FRW cosmology:

$$\ddot{G} + 3H\dot{G} + m_G^2 G = 0. \quad (8)$$

4 Stress–Energy Tensor and Cosmology

The stress–energy tensor is:

$$T_{\mu\nu}^{(G)} = \nabla_\mu G \nabla_\nu G - g_{\mu\nu} \left(\frac{1}{2} (\nabla G)^2 + \frac{1}{2} m_G^2 G^2 \right). \quad (9)$$

For homogeneous $G(t)$:

$$\rho_G = \frac{1}{2} \dot{G}^2 + \frac{1}{2} m_G^2 G^2, \quad p_G = \frac{1}{2} \dot{G}^2 - \frac{1}{2} m_G^2 G^2. \quad (10)$$

The equation-of-state parameter is:

$$w = \frac{p_G}{\rho_G}. \quad (11)$$

DESI measurements imply:

$$\frac{K}{V} = \frac{\dot{G}^2}{m_G^2 G^2} = 0.0811, \quad (12)$$

consistent with a slowly rolling scalar field.

5 Running Gravitational Constant and Collapse Reversal

We define the effective gravitational coupling:

$$G_{\text{eff}} = G_0(1 - \gamma G). \quad (13)$$

Collapse halts when:

$$G(M_{\text{freeze}}) = \gamma^{-1}, \quad (14)$$

yielding stable relics with finite radius and constant energy density.

6 Forward–Backward Consistency Demonstration

Choose a concrete scalar field:

$$G(t) = 10^{-3} + 10^{-4} \cos(\omega t). \quad (15)$$

Forward (decay sector):

$$\frac{\Delta\lambda(t)}{\lambda_0} = 10^{-3} + 10^{-4} \cos(\omega t). \quad (16)$$

Backward reconstruction:

$$G_{\text{recon}}(t) = \frac{\Delta\lambda(t)}{\alpha\lambda_0} = 10^{-3} + 10^{-4} \cos(\omega t). \quad (17)$$

Thus:

$$G_{\text{recon}}(t) = G(t), \quad (18)$$

demonstrating exact forward–backward consistency.

7 Empirical Anchoring

The amplitude $G \sim 10^{-3}$ matches:

- Mn-54 reactor shutdown anomalies,
- Si-32 and Cl-36 drift data,
- PTB decay-rate datasets,
- lunar MASCON gravitational anomalies,
- Apollo inertial residuals,
- DESI cosmological scalar-field ratio,
- relic freeze conditions,
- laboratory transient predictions.

8 Conclusion

A single scalar field $G(t)$ with amplitude $G \sim 10^{-3}$ explains cross-sector anomalies through a unified correction law. The theory is dynamically complete, cosmologically consistent, and empirically anchored. Forward–backward consistency demonstrates that $G(t)$ is a measurable physical field with predictive and reconstructive power across nuclear, gravitational, inertial, and cosmological domains.